

Investigating motor skill training and user arousal levels in VR : Pilot Study and Observations

Unnikrishnan Radhakrishnan*
Department of Business
Development and Technology
Aarhus University

Alin Blindu†
Aarhus University

Francesco Chinello‡
Department of Business
Development and Technology
Aarhus University

Konstantinos Koumaditis §
Department of Business
Development and Technology
Aarhus University

ABSTRACT

Virtual Reality (VR) for skill training is seeing increasing interest from academia and industry thanks to the highly immersive and realistic training opportunities they offer. Of the many factors affecting the effectiveness of training in VR, the arousal levels of the users merit a closer examination. Though subjective methods of measuring arousal exist in VR literature, there is potential in using cost-effective sensors to directly measure these from bio-signals generated by the nervous system. We introduce the design of preliminary observations from a pilot study exploring user's arousal levels and performance while executing a series of fine motor skill tasks (buzzwire tracing). Future directions of the work are also discussed.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual Reality;

1 INTRODUCTION

VR skill training systems offer several affordances such as providing familiarity with the associated real-world environment, increasing the salience of particular cues in the environment, allowing learners to rehearse specific sequences of actions with a high level of physical and psychological fidelity, and enabling immediate feedback and broad accessibility in a safe and controlled environment [3] [4]. Though VR based skill training is increasing in popularity across various domains, there is a need for a closer examination of the link between the user's arousal levels and training performance. Findings from such investigations could potentially be used to design adaptive VR training systems [2]. Compared to post-exposure subjective measures of arousal which rely on the user's memory of the task and comprehension of questions, objective measures of arousal are a function of the body's autonomic nervous system, which produces measurable responses reflecting the user's mental state. This includes changes in skin conductance (due to sweating) and heart rate parameters, among others which can be measured by various off-the-shelf sensors. Objective bio-signal data also allows for a more fine-grained look at variations in arousal across tasks with varying difficulty levels.

2 EXPERIMENTAL SETUP

The buzzwire levels proposed were replicated both in VR and physical environments. The physical setup consisted of metal wires bent to fit the design of the different levels (see Fig. 1) while the VR setup is visualised in a VR environment using the Unity3D game engine and Oculus SDK. In the physical condition, the user is in a seated position with the dominant hand holding the Oculus controller enclosed in a 3D printed handle [5] coupled to the circular ring (of

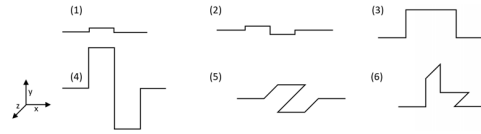


Figure 1: Buzzwire tracing task levels with (1) to (4) in the xy plane, (5) in the xz plane and (6) in all three axes.

diameter 2.5 cm) as in Fig. 2(d). In the VR scenario, the user is in a seated position wearing an Oculus Quest and has one Oculus controller in the dominant hand.

The proposed buzzwire tracing task involves the user moving the circular ring across a thin wire (as in Fig. 2(a)), without touching it. The wire is bent at certain locations providing the challenge of maintaining proper hand-eye coordination while executing the task. This scenario has been used previously in VR and non-VR training scenarios [1] [6]. In order to explore the variation in arousal levels to the difficulty levels of the task, 6 difficulty levels were designed (see Fig. 1) to select a few representative levels for future experiments with larger populations. The total lengths of all 6 levels were kept at a constant 48cm.

The levels are designed as follows: level 1 has a bend with a height of 1.25 cm so that the user can pass the ring through x-axis (see Fig. 1); levels 2, 3 and 4 require hand rotations between the x and y axes. Level 5 changes the plane of motion (from xy to xz) while level 6 requires movement in all three axes.

An Arduino-based program detects when the ring touches the wire, and immediately switches on a red LED (Fig. 2(a)) to glow while simultaneously transmitting the data to the PC where a Unity3D script reads the data and triggers a short vibration (via the Oculus controller) and a sharp tone (via a Bluetooth-headphone) for each error. In the VR condition, Unity3D mesh-colliders are used to detect collisions between the ring and wire. For the VR setup, the audio and haptic feedback is the same as the physical condition, but for the visual feedback, a virtual light source in Unity3D is created which emits red light.

2.1 Bio-signal and Performance Measures

A Shimmer GSR+ (Galvanic Skin Resistance) unit (Fig. 2(c)) on the non-dominant hand (resting on the table) is used for measuring skin conductance (SC), and the Polar H10 chest strap is used for measuring the heart rate. Though SC can be recorded from any part of the skin, certain locations on the body are reported by literature to be better for recording higher quality data [7]. This was verified during a pretest comparing the quality of signals from different locations on the body, among which the fingers were identified after testing as being the most reliable. Ag/AgCl electrodes are placed on the index and middle fingers of the non-dominant hand, while the dominant hand is used to move the ring around the wire. The non-dominant hand is kept stationary throughout the experiment as GSR is affected by motion artifacts. Timestamped data from both Shimmer and Polar devices were collected by the iMotions software for later processing and analysis. Established measures of buzzwire

*e-mail: unnik@btech.au.dk

†e-mail: alinblindu@gmail.com

‡e-mail: chinello@btech.au.dk

§e-mail: kkoumaditis@btech.au.dk

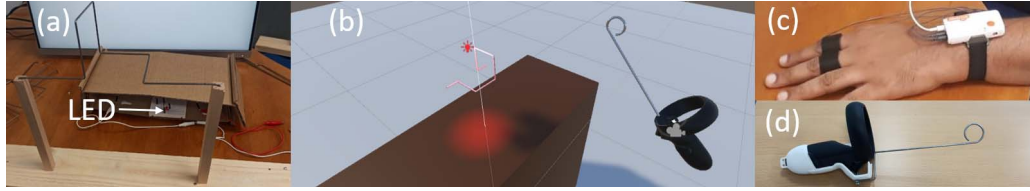


Figure 2: (a) Physical setup of level 6, (b) VR setup of level 6, (c) Shimmer GSR+ sensor, (d) Oculus controller with ring attached.

tracing (and similar fine motor skill tasks) like task completion time and count of errors are generated based on the timestamped data from iMotions during the data processing stage.

3 PRELIMINARY RESULTS AND ANALYSIS

Twelve participants volunteered to be part of the pilot study, 6 in the physical condition and 6 in the virtual reality condition, with three trials per level. The physical experimentation was performed with the biosensors, but the VR experiment was affected by Covid-19 restrictions and volunteers had access to a VR headset in an open area. The number of mistakes and task completion time were measured for both conditions.

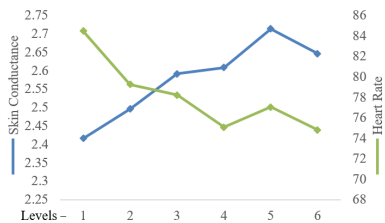


Figure 3: Average Skin Conductance (μS) & Average Heart Rate data across levels 1 to 6 (physical condition).

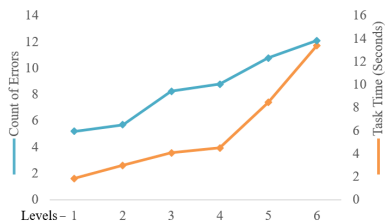


Figure 4: Average Count of Errors & Task Time (seconds) across levels 1 to 6 (physical condition)

Fig. 3 summarizes the results from the physical training tests. SC levels continue to climb on average for the 6 participants, though there is an unexpected drop from level 5 to level 6. HR drops on average from levels 1 to 6, though there is a slight climb from level 4 to 5 beating the downward trend. On the other hand, as shown in Fig. 4, task completion time and count of errors show a steady climb from levels 1 to 6, indicating that the initial reasoning for the design of the difficulty levels in the levels is sound. The climb observed in bio-sensor data for level 5 might be due to the novelty and challenge of tracing in a different plane compared to the previous levels. The data from the 6 participants in the VR condition (count of errors and task completion time) also confirmed these observations with a steady increase from levels 1 to 6.

Interviews and observations of both the physical and VR pilots suggest constructive changes and further research questions be ex-

plored. These include the role of decision making in the learning of fine motor skills since more than one participant mentioned the cognitive challenge of finding optimal solutions to levels 5 and 6. Though the buzzwires and the handle for the ring are similar in size across VR and physical conditions, there were differences in other aspects of the visual environment (table, background, etc). Some participants mentioned that they felt a slight delay in getting audio feedback when they commit a mistake, which is most possibly due to the Bluetooth connection. Two participants mentioned that they did not pay much attention to the visual feedback (LED) while for one person that was the primary modality.

4 LESSONS LEARNED AND FUTURE WORK

The pilot study showed that (a) the bio-sensor setup (Shimmer GSR+ on the fingers and Polar H10 on the chest) generates reliable data, (b) the SC levels scaled up with the difficulty levels while the HR levels showed the opposite trend (c) the position of the LED (visual feedback) may have to be changed to a more prominent position as it was ignored by certain participants (d) the VR and physical experimental setups need to be designed to match each other closely as possible. Further investigation with a larger sample of participants needs to be done to conclude the relation between arousal levels as measured by the bio-sensors and training performance. The pilot study also sets the stage for robust experiments in the future to explore the role of audio/visual/haptic feedback (by comparing to the absence of it) and biofeedback (by allowing the user to see some visualization of their bio-sensor data) in affecting training performance across VR and physical conditions. More fine-grained analysis of bio-sensor data will also be explored since the pilot study only analyzed the skin conductance levels and not the skin conductance responses (peaks) that follow stimuli (in this case, errors and feedback) [7].

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